Updating an Item in Azure Cosmos DB Using the .NET SDK

Summary:-

In this lab, you will update an existing document in an Azure Cosmos DB container using the Azure Cosmos DB .NET SDK. You will retrieve an item using its **Item ID** and **Partition Key**, modify one of its properties, and replace the existing document in the container.

Let's Start:-

After completing this lab, you will be able to:

- Read an existing document from Azure Cosmos DB.
- Modify the properties of a retrieved document.
- Update an item using the ReplaceItemAsync() method.
- Verify the updated document in Azure Cosmos DB.

Lab Steps:-

Step 1 – Prepare the Azure Cosmos DB .NET Project

- **Reference:** This lab uses the .NET console application, Azure Cosmos DB account, database, and container created in the previous lab.
- For that **create** Azure Cosmos DB account by selecting **Azure Cosmos DB for NoSQL** from the recommended APIs.
- Configure:
  Workload type: Learning, **Region:** EAST US/ EAST US 2, **Capacity Mode:** Provisioned Throughput
- Enable **Apply Free Tier Discount.** Click **Next**.
- Ensure Geo-Redundancy and Multi-region Writes is **Disabled**. Click **Next**.
- Select **Enable access from all networks**. Click on **Review + Create** to create.
- From the left pane, select **Keys**.
- Copy: **URI** and **Primary Key**
- Open **Visual Studio Code**. Create a new project folder.
- Press **Ctrl + Shift + P** to open the **Command Palette**.
- Select **Show and Run Commands** > **.NET: New Project** > **Console App**(Rename it as cosmosdb-app) > **Default Directory** > **.slnx** > **Create Project**.
- **Open** the integrated terminal. Run:

```
cd .\cosmosdb-app\
dotnet add package Microsoft.Azure.Cosmos
dotnet add package Newtonsoft.Json
```

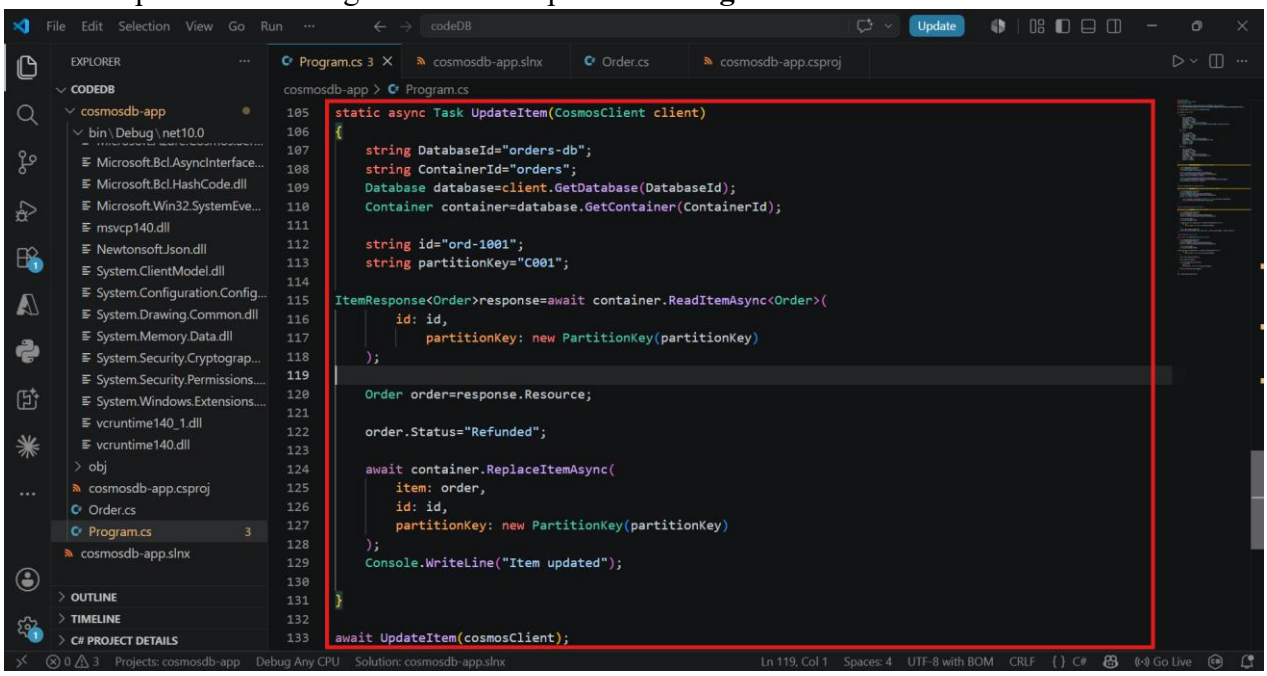
- Create a new class file named **Order.cs**.
- Open the following project files:

Program.cs
Order.cs
cosmosdb-app.csproj
cosmosdb-app.slnx

- Replace the existing content with the **provided files**.
- In Program**.cs, Replace** Endpoint with **URI** and Key with **Primary Key**.
- **Save** all the files. **Run**:
  dotnet run
  dotnet build
- This step creates a .NET console application and prepares it to insert items into the Azure Cosmos DB container.

Step 2 – Add the Update Item Code

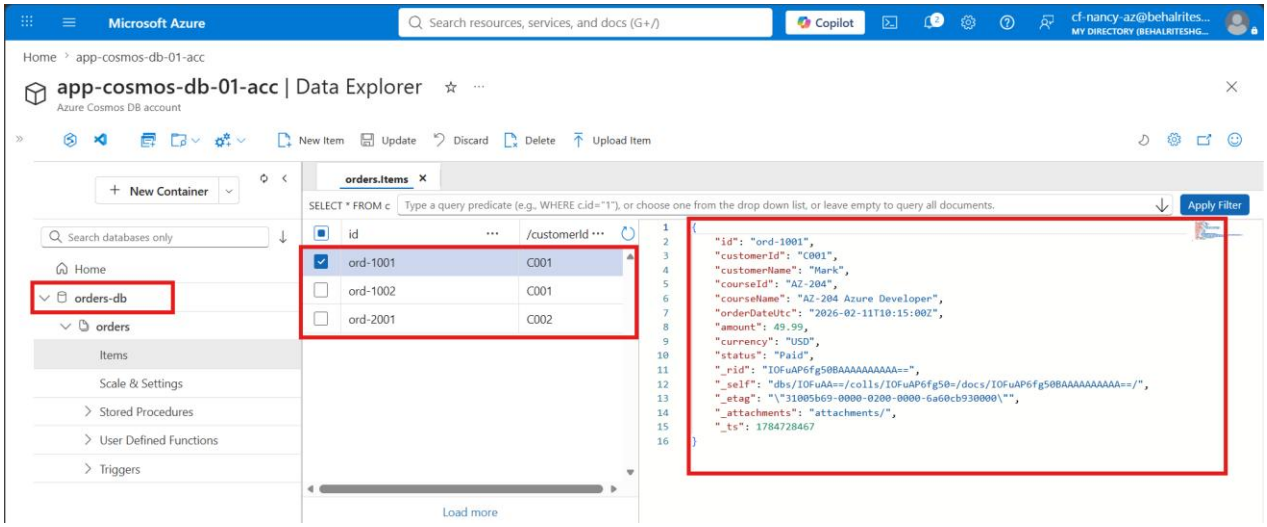
- Open **Program.cs**.
- Replace the existing code with the provided **Program.cs** file.



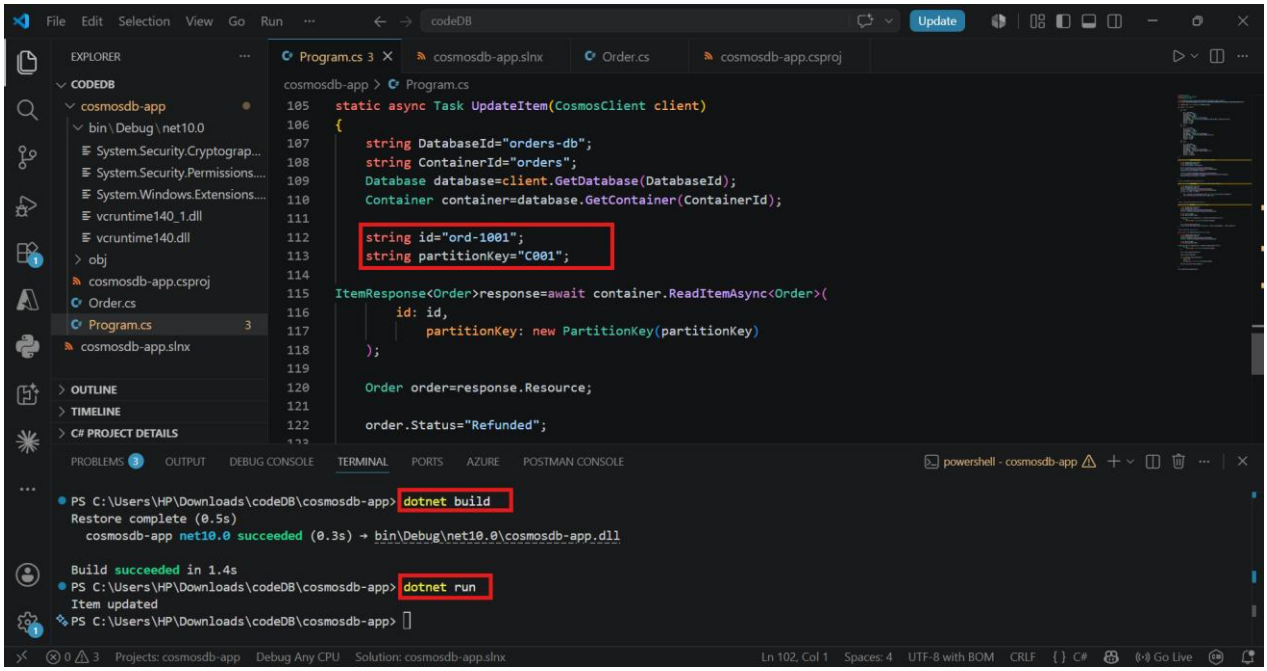
- **Save** the file.
- This step adds the code required to update an existing document in Azure Cosmos DB.

Step 3 – Update the Item

- In the code, specify:
  - **Item ID:** ord-1001
  - **Partition Key:** C001
- Observe that the **Status** field is currently set to **"Paid"** before running the application.



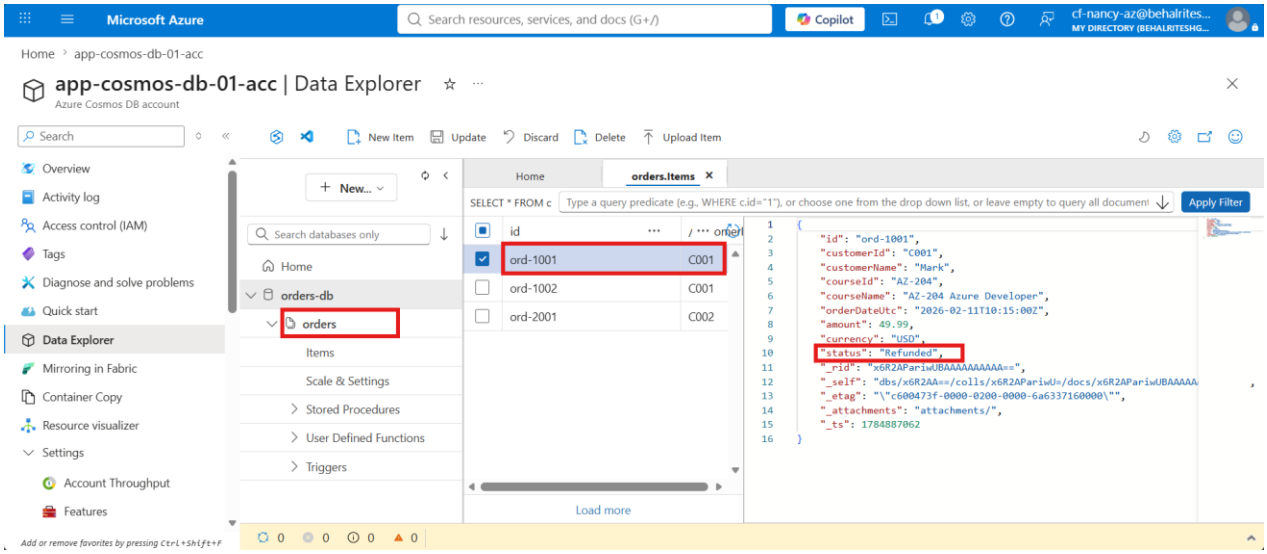
- **Build** and **run** the application. Run:
  dotnet build
  dotnet run



- This step reads the existing document, updates its status, and replaces the document in the container.

Step 4 – View the Retrieved Item

- Open the **Azure Portal**. Navigate to **Azure Cosmos DB** > **Data Explorer**.
- Open the **orders** container.
- Select the updated document (ord-1001).



- Verify that the **Status** field has been updated successfully.
- This step confirms that the document has been updated successfully in Azure Cosmos DB.

Verification

Verify the following:

- The application runs successfully without errors.
- The specified document is retrieved correctly.
- The **Status** field is updated successfully.
- The updated document is visible in the Azure Cosmos DB container.
- The changes are saved successfully using the **ReplaceItemAsync()** method.